Proofs about Linear Maps

Existence, uniqueness

MTH 317H · Week 7 · Review + proof writing

Name: _____

Today at a glance

By the end of class you should be able to:

1. pick the right *proof framework* (first line, last line) for "injective", "surjective", "not injective", "not surjective";
2. compute $\operatorname{null} T$ and $\operatorname{range} T$ and use them to decide injectivity and surjectivity;
3. spot (and repair) the most common logical errors in proofs about functions;
4. find and verify inverses, and prove facts about compositions.

1 Warm-up: sort the maps

Here are four linear maps.

$$D\colon \mathcal{P}_3 \to \mathcal{P}_3,\quad D(p) = p', \qquad S\colon \mathbb{R}^2 \to \mathbb{R}^3,\quad S(x,y) = (x,\ y,\ x+y),$$
$$P\colon \mathbb{R}^3 \to \mathbb{R}^2,\quad P(x,y,z) = (x,\ y), \qquad E\colon \mathcal{P}_2 \to \mathbb{R}^3,\quad E(p) = \big(p(0),\ p(1),\ p(2)\big).$$

Task 1.1 (Gut feeling first)**.** Place each map in the grid with a *one-line reason.*

	injective	**not injective**
surjective		
not surjective		

Task 1.2 (Can you fill every cell?)**.** (a) Invent a linear map $\mathbb{R}^4 \to \mathbb{R}^3$ for each cell of the grid, if you can. Which cells seem impossible?

(b) Do the same for linear maps $\mathbb{R}^4 \to \mathcal{P}_3$.

Write down a conjecture. *We will come back to it in Task 4.4.*

2 Notes: two questions about every equation $T(v) = w$

Almost every question this week is a question about solving one equation

$$T(v) = w, \qquad T\colon V \to W \text{ given}, \quad w \in W \text{ a target}, \quad v \in V \text{ unknown}.$$

For a given target there are always two questions: **Existence**: is there *at least one* solution? **Uniqueness**: is there *at most one*?

Notes: definitions

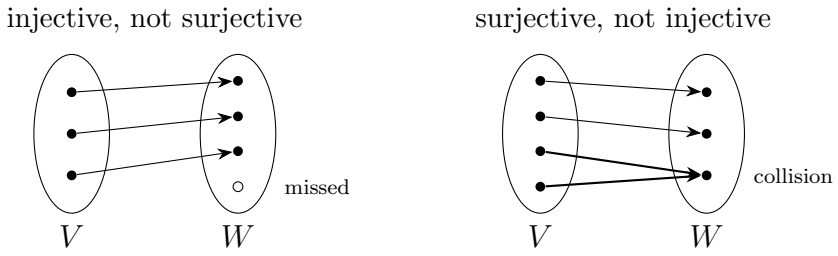
Let $T\colon V \to W$ be a function.

- T is **surjective** (onto) if for every $w \in W$ there *exists* $v \in V$ with $T(v) = w$. *(existence, for every target)*
- T is **injective** (one-to-one) if $T(u) = T(v)$ implies $u = v$. *(uniqueness, for every target)*
- T is **bijective** if it is both, so each target has *exactly one* solution. Then "solve $T(v) = w$" defines a function $W \to V$: the **inverse** T^{-1}.

If T is linear: $\operatorname{null} T = \{v \in V : T(v) = 0\}$ and $\operatorname{range} T = \{T(v) : v \in V\}$.

Key fact. A linear map T is injective $\iff \operatorname{null} T = \{0\}$. *Why:* by linearity, $T(u) = T(v) \iff T(u - v) =$ _____ $\iff u - v \in$ _____.

So for linear maps, uniqueness for *every* target can be checked at *one* target, $w = 0$. That is the payoff of linearity.



Notes: proof frameworks (write the first and last lines *before* you think)

The first and last lines of a proof come straight from the definitions. Write them down first, every time. The real problem-solving happens in the middle.

To prove ...	**First line**	**Last line**
T linear	Let $u, v \in V$ and $\lambda \in \mathbb{R}$.	Therefore $T(u+v) = T(u) + T(v)$ and $T(\lambda u) = \lambda T(u)$.
T *not* linear	Exhibit specific u, v (or λ, u) ...	... for which additivity (or homogeneity) fails. Quick test: if $T(0) \neq 0$, then T is not linear.
T injective (definition)	Let $u, v \in V$ and suppose $T(u) = T(v)$.	Therefore $u = v$.
T injective (T linear)	Let $v \in \operatorname{null} T$, i.e. $T(v) = 0$.	Therefore _____.
T surjective	_____ *(scratch work: solve $T(v) = w$, hidden from the final proof)* Define $v := \ldots$	Then $T(v) = \cdots = w$, so $w \in \operatorname{range} T$.
T *not* injective	Exhibit specific $u \neq v$ ...	... with $T(u) = T(v)$. One example is enough.
T *not* surjective	Exhibit a specific $w \in W$ ...	... and prove $T(v) \neq w$ for _____ $v \in V$.

Connection to Lecture 7 (bases)

- Recall the proof in Lecture 7 that $(1,2,0), (2,0,0), (3,1,2)$ span $\mathbb{R}^3$: *exploration* (solve for a_1, a_2, a_3) → "Let $(x,y,z) \in \mathbb{R}^3$ be given, arbitrary" → "Define $a_1 = \ldots$" → "Then $a_1v_1 + a_2v_2 + a_3v_3 = (x,y,z)$." That is *exactly* the surjectivity framework.
- The proposition "$v_1, \ldots, v_n$ is a basis $\iff$ every $v \in V$ can be written *uniquely* as a linear combination" is existence + uniqueness in disguise.

The dictionary. Given a list $v_1, \ldots, v_n$ in V, define $S\colon \mathbb{R}^n \to V$ by $S(a_1, \ldots, a_n) = a_1v_1 + \cdots + a_nv_n$. Then

the list $v_1, \ldots, v_n$ ...	**the map S ...**	**the question**
spans V	is surjective	existence of coefficients
is linearly independent	is injective	uniqueness of coefficients
is a basis of V	is bijective	exactly one choice of coefficients

(You can prove this in Task 3.8 ★.)

3 Investigation: encoding $\mathbb{R}^4$ as cubic polynomials

Riley wants to send four numbers (a, b, c, d) to Jordan, but the only channel available transmits cubic polynomials. Riley encodes the numbers with the linear map $T\colon \mathbb{R}^4 \to \mathcal{P}_3$,

$$T\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = bx^3 + (c+d)x^2 + (a-b)x - (a+d).$$

(We write (a, b, c, d) for the column vector when it fits better in a line of text.) Jordan receives a polynomial and has to *decode* it. Two questions drive this section:

- Can Jordan always decode *unambiguously*? (Is T injective?)
- Is *every* cubic polynomial a legitimate message? (Is T surjective?)

Task 3.1 (Get to know T)**.** Compute $T(e_1), T(e_2), T(e_3), T(e_4)$ and $T(1, 1, 1, 1)$. How is $T(1, 1, 1, 1)$ related to the first four? Write $T(a, b, c, d)$ as a linear combination of four *fixed* polynomials.

Task 3.2 (Twin hunt)**.** We know $T(1, 1, 1, 1) = x^3 + 2x^2 - 2$. Try to find a *different* $v \in \mathbb{R}^4$ with $T(v) = x^3 + 2x^2 - 2$, a "twin" of $(1, 1, 1, 1)$. Record what you tried. If you cannot find one, does that prove there is none? What exactly would a proof have to show?

Task 3.3 (Prove it: injectivity from the definition)**.** Complete the proof. The first and last lines are given.

Proof. Let $u = (a, b, c, d)$ and $u' = (a', b', c', d')$ be in $\mathbb{R}^4$, and suppose $T(u) = T(u')$, i.e.

$$bx^3 + (c+d)x^2 + (a-b)x - (a+d) \;=\; b'x^3 + (c'+d')x^2 + (a'-b')x - (a'+d').$$

Two polynomials in $\mathcal{P}_3$ are equal exactly when _____ (this uses the linear independence of $1, x, x^2, x^3$). Comparing coefficients,

Therefore $u = u'$, and T is injective. □

Task 3.4 (Prove it again: the linear shortcut)**.** Find $\operatorname{null} T$ and conclude that T is injective. Then:

(a) Compare with Task 3.3. How is this system of equations related to the one there?

(b) Where *exactly* is linearity used when you deduce injectivity from $\operatorname{null} T = \{0\}$?

(c) Finish the twin hunt: in one sentence, explain why $(1, 1, 1, 1)$ has no twin.

Task 3.5 (Decode a message)**.** Jordan receives $2x^3 + x^2 - 3x - 1$. Find the message $v \in \mathbb{R}^4$, and *check* your answer by computing $T(v)$.

Task 3.6 (Decode *every* message: surjectivity)**.** Prove that T is surjective, following the framework.

Scratch work (not part of the final proof). Given $q = \alpha x^3 + \beta x^2 + \gamma x + \delta$, solve $T(a, b, c, d) = q$ for a, b, c, d in terms of $\alpha, \beta, \gamma, \delta$.

Proof. Let $q = \alpha x^3 + \beta x^2 + \gamma x + \delta \in \mathcal{P}_3$ be arbitrary. Define $v = \big(_____, _____, _____, _____\big)$. Then $T(v) = \ldots$ (compute):

Hence $T(v) = q$. Since q was arbitrary, T is surjective. □

Afterwards: your formula $q \mapsto v(q)$ is a *decoder*, a function $\mathcal{P}_3 \to \mathbb{R}^4$. Test it on the message of Task 3.5. What are $T(v(q))$ and $v(T(a, b, c, d))$? (Keep this in mind for Section 6.)

Task 3.7 (What if not? Design a leaky encoder)**.** (a) Change the formula for T *a little* (one letter or one sign) so that the new map T' is *not* injective. Find a *collision*: two different messages with the same code.

(b) Is your T' surjective? Prove it, or find a polynomial that no message encodes to.

(c) Use your example to conjecture a relationship between injectivity and surjectivity for a linear map $\mathbb{R}^4 \to \mathcal{P}_3$.

Task 3.8 (★ The dictionary)**.** Let $v_1, \dots, v_n \in V$ and $S(a_1, \dots, a_n) = a_1v_1 + \cdots + a_nv_n$. Prove: (a) S is linear; (b) S is injective $\iff$ $v_1, \dots, v_n$ is linearly independent; (c) S is surjective $\iff$ $v_1, \dots, v_n$ spans V. (d) Use Task 3.1 to conclude that $x - 1,\ x^3 - x,\ x^2,\ x^2 - 1$ is a basis of $\mathcal{P}_3$.

Takeaways

- For a linear map, to prove uniqueness for *every* target it is enough to solve $T(v) =$ _____.
- A surjectivity proof has three moves: take an _____ target; *define* a candidate (found by scratch work); _____ that it maps to the target.
- Not finding a counterexample is _____, not _____.

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4 Investigation: squeezing $\mathbb{R}^4$ into $\mathbb{R}^3$

Now four numbers are compressed into three:

$$T\colon \mathbb{R}^4 \to \mathbb{R}^3, \qquad T\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 2x + z - 3t \\ y - 7t \\ 6x - y + 3z - 2t \end{pmatrix}.$$

Some information has to be lost. *Where does it go, and what gets through?*

Task 4.1 (Predict, then check linearity)**.** (a) Before computing anything: could T be injective? Could it be surjective?

(b) Prove that T is linear. *Hint:* write $T(x, y, z, t) = x(\ ) + y(\ ) + z(\ ) + t(\ )$.

(c) *Pose:* change the formula slightly so that T becomes *non*-linear, and give a specific counterexample showing that additivity or homogeneity fails.

Task 4.2 (What gets lost: $\operatorname{null} T$)**.** (a) Find $\operatorname{null} T$ and write it as the span of a list of vectors.

(b) Is your list a basis of $\operatorname{null} T$? Justify.

(c) Use (a) to produce a *collision*: two different inputs with the same output.

(d) Find *all* v with $T(v) = T(1, 1, 1, 1)$. Compare with the twin hunt in Section 3.

Task 4.3 (What gets through: $\operatorname{range} T$)**.** (a) Explain why $\operatorname{range} T = \operatorname{span}\big(T(e_1), T(e_2), T(e_3), T(e_4)\big)$, and compute these vectors.

(b) *The checksum.* Find numbers p, q such that *every* output $(w_1, w_2, w_3) = T(v)$ satisfies $w_3 = p\,w_1 + q\,w_2$. (Compare the third formula with the first two.)

(c) Prove $\operatorname{range} T = \{(w_1, w_2, w_3) \in \mathbb{R}^3 : w_3 = 3w_1 - w_2\}$. (Two inclusions!)

(d) Prove that T is not surjective. Your argument must rule out *every* input.

Task 4.4 (From data to a conjecture)**.** Complete the table, using Sections 1, 3 and 4 (for the warm-up maps, rely on your answers to Task 1.1).

map	$\dim V$	$\dim \operatorname{null} T$	$\dim \operatorname{range} T$	inj.?	surj.?
$T\colon \mathbb{R}^4 \to \mathcal{P}_3$ (Section 3)	____	____	____	____	____
$T\colon \mathbb{R}^4 \to \mathbb{R}^3$ (Section 4)	____	____	____	____	____
$D\colon \mathcal{P}_3 \to \mathcal{P}_3$	____	____	____	____	____
$S\colon \mathbb{R}^2 \to \mathbb{R}^3$	____	____	____	____	____
$P\colon \mathbb{R}^3 \to \mathbb{R}^2$	____	____	____	____	____
$E\colon \mathcal{P}_2 \to \mathbb{R}^3$	____	____	____	____	____

(a) Conjecture a formula relating $\dim V$, $\dim \operatorname{null} T$ and $\dim \operatorname{range} T$.

(b) Assuming your formula, explain the "impossible cells" from Task 1.2.

Homework launch: HW 5

The next two tasks get you started on HW 5. They help you plan, but they do not do the homework for you.

Task 4.5 (Null spaces in disguise)**.** Many subspaces are cut out by linear equations, which makes them null spaces. For example, the subspace from HW 5, Problem 4,

$$U = \{x \in \mathbb{R}^5 : x_1 = 2x_2 \text{ and } x_4 = 6x_5\},$$

is $\operatorname{null} L$ for the linear map $L\colon \mathbb{R}^5 \to \mathbb{R}^2,\ L(x) = (x_1 - 2x_2,\ x_4 - 6x_5)$.

(a) Check that $U = \operatorname{null} L$ and that L is surjective. Assuming the formula from Task 4.4, what must $\dim U$ be?

(b) Do the same for HW 5, Problem 5: $U = \{p \in \mathcal{P}_3 : p'(2) = 0\}$. Find a linear map $\varphi\colon \mathcal{P}_3 \to \mathbb{R}$ with $\operatorname{null} \varphi = U$. What must $\dim U$ be?

(c) How does knowing $\dim U$ in advance help you check your homework answers? *Hint:* in LADR 2.C, what can you say about a linearly independent list in U whose length is $\dim U$?

Task 4.6 (Linear or not?)**.** For each map, predict whether it is linear, then *plan* the proof: write its first line from the framework table, or, if you think the map is not linear, the specific inputs you would use as a counterexample. Do not write the full proofs now; that is the homework.

map (HW 5, Problem 8)	**linear?**	**your plan**
(i) $\mathcal{P}_4 \to \mathcal{P}_4,\ p \mapsto x^2 p''(x)$		
(ii) $\mathcal{P}_4 \to \mathcal{P}_4,\ p \mapsto p''(x) + x^2$		
(iii) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 x_2,\ x_1,\ 5x_3 + x_4)$		
(iv) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 - 3x_2,\ x_1 + 7,\ 5x_3 + x_4)$		
(v) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 - 3x_2,\ x_1,\ 5x_3 + x_4)$		

5 Grade the proof

Each argument below was submitted by a (fictional) student. For each one: **(1)** Verdict: is it a valid proof? **(2)** If not, point to the *first* line that fails and say why. **(3)** Repair it, or say that the claim itself is false. Read line by line and ask, every time, "*Why does this follow?*"

A. *Claim: T from Section 3 is injective.* "Since T is linear, $T(0) = 0$. So $0 \in \operatorname{null} T$, and therefore $\operatorname{null} T = \{0\}$. Hence T is injective."

B. *Claim: T from Section 3 is surjective.* "Let $v = (a, b, c, d) \in \mathbb{R}^4$. Then $T(v) = bx^3 + (c+d)x^2 + (a-b)x - (a+d)$ has degree at most 3, so $T(v) \in \mathcal{P}_3$. Since v was arbitrary, T is surjective."

C. *Claim: T from Section 3 is injective.* "Let $u, u' \in \mathbb{R}^4$ with $u = u'$. Then $T(u) = T(u')$, since T is a function. Hence T is injective."

D. *Claim: T from Section 4 is not surjective.* "$T(e_1) = (2, 0, 6)$, $T(e_2) = (0, 1, -1)$, $T(e_3) = (1, 0, 3)$, $T(e_4) = (-3, -7, -2)$. None of these is $(0, 0, 1)$, so $(0, 0, 1) \notin \operatorname{range} T$. Hence T is not surjective."

E. *Claim: T from Section 4 is not injective.* "$T(1, 0, -2, 0) = (2 - 2,\ 0,\ 6 - 6) = (0, 0, 0) = T(0, 0, 0, 0)$, and $(1, 0, -2, 0) \neq (0, 0, 0, 0)$. So T is not injective."

Task 5.1 (★ Turn the tables)**.** Write a short "proof" about one of today's maps that *looks* convincing but contains one subtle error. Identify the error and explain how to correct it.

Classic traps

1. $T(0) = 0$ holds for *every* linear map. It says $0 \in \operatorname{null} T$, not $\operatorname{null} T = \{0\}$.
2. Mind the quantifier order. Surjective means *for every w there is a v*, so start from w.
3. Don't prove the converse. Injective is $T(u) = T(u') \Rightarrow u = u'$, not the other way around.
4. Finitely many checks can *disprove* a "for all" statement, but they can never prove a "there is no ..." statement.

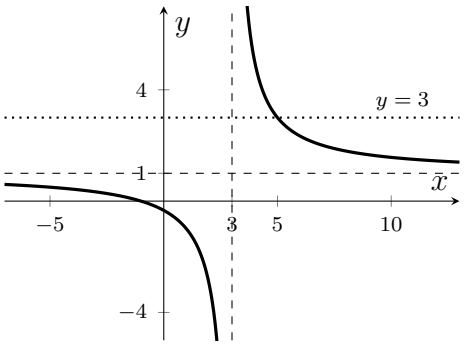
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6 Inverses and compositions (Week 6, Problems 5 and 7)

In Task 3.6 you built a decoder that undoes T. In Week 6 you proved that $f\colon X \to Y$ *is bijective if and only if it has an inverse* $h\colon Y \to X$, meaning $h(f(x)) = x$ for all $x \in X$ and $f(h(y)) = y$ for all $y \in Y$. The two problems below involve functions that need not be linear. The frameworks from Section 2 still apply word for word.

An inverse with excluded values

$$r\colon \mathbb{R} \setminus \{3\} \longrightarrow \mathbb{R} \setminus \{1\}, \qquad r(x) = \frac{x+1}{x-3}.$$



Task 6.1 (Look before you prove)**.** (a) Which horizontal lines $y = c$ meet the graph exactly once? Which meet it not at all?

(b) What does this "horizontal line test" have to do with injectivity and surjectivity?

(c) Why is the codomain $\mathbb{R} \setminus \{1\}$ rather than $\mathbb{R}$?

Task 6.2 (Prove it)**.** (a) Check that the formula defines a function $\mathbb{R} \setminus \{3\} \to \mathbb{R} \setminus \{1\}$. There are *two* things to check.

(b) *Scratch work:* solve $y = r(x)$ for x. Use the result to define a candidate inverse s, stating its domain and codomain.

(c) Check that s takes values in its claimed codomain, and verify both compositions. Conclude that r is bijective.

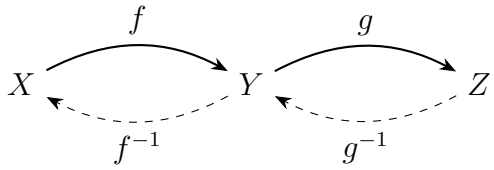
Task 6.3 (What if not?)**.** (a) Keep the formula, but change the codomain to $\mathbb{R}$. Which of injective, surjective, bijective survive?

(b) For $r_2(x) = \dfrac{2x+1}{x-3}$, which value must be removed from the codomain? Conjecture the excluded output for $\dfrac{ax+b}{cx+d}$ with $c \neq 0$.

(c) ★ Try $\dfrac{2x+6}{x+3}$. What goes wrong? Find a condition on a, b, c, d that rules this out.

(d) ★ Write the coefficients of r and of your s as 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and multiply them. Notice anything?

Composition of bijections



Task 6.4 (Socks and shoes)**.** (a) *Warm-up.* Let $f, g\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x + 1$, $g(x) = 2x$. Compute $g \circ f$ and $(g \circ f)^{-1}$. Which one equals $(g \circ f)^{-1}$: $g^{-1} \circ f^{-1}$ or $f^{-1} \circ g^{-1}$? Explain using socks and shoes.

(b) Now let $f\colon X \to Y$ and $g\colon Y \to Z$ be bijective. Prove directly that $g \circ f$ is injective. *(Which row of the framework table are you in?)*

(c) Prove directly that $g \circ f$ is surjective.

(d) Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by checking both compositions.

Task 6.5 (What if not? Converses)**.** Suppose we know only that $g \circ f$ is bijective. Must f be injective? Surjective? Must g be injective? Surjective? Make four conjectures, then prove each one or find a counterexample using small finite sets and arrow diagrams.

Task 6.6 (★ Back to linear maps)**.** Let $S\colon U \to V$ and $T\colon V \to W$ be linear and injective. Give a two-line proof, using null spaces, that $T \circ S$ is injective.

7 Exit ticket (tear off and hand in)

Name: _____

1. To prove that $T\colon V \to W$ is *surjective*, my first line is _____ and my last line is _____

2. To prove that T is *not* surjective, I must

   _____

3. Rate your confidence, 1 (lost) to 4 (I could teach it):

proving a map is injective	1	2	3	4
proving a map is surjective	1	2	3	4
computing $\operatorname{null} T$ and $\operatorname{range} T$	1	2	3	4
finding and verifying an inverse	1	2	3	4

4. **Pose a problem.** Write one question about today's ideas that you would like answered, *or* a problem you would put on the exam.